

COMPLETE LEARNING SESSION

Number-Systems-and-Number-Sense - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Number-Systems-and-Number-Sense - Complete CSAT Learning Session

Identity: csat-02: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** Basic numeracy: numbers and their relations, orders of magnitude, etc. (Class X level).

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The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

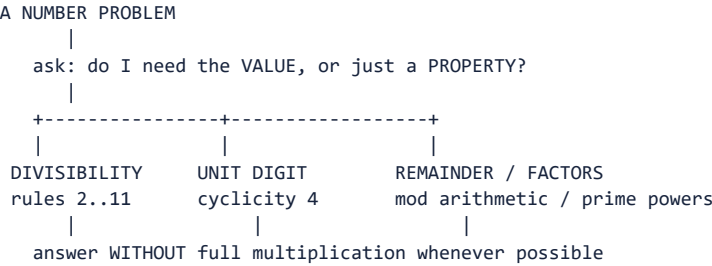
Twelve-panel ASCII master flow

	PANEL 01 - NUMBER UNIVERSE	
	Natural, whole, integer, rational, irrational and real numbers.	
+		+
	v	
+		+
	PANEL 02 - DIVISIBILITY	
	Use prime-factor and digit tests only within their valid base-ten	
	conditions.	
+		+
	v	
+		+
	PANEL 03 - PRIMES AND FACTORS	
	Factorise before counting divisors or comparing powers.	
+		+
	v	
+		+
	PANEL 04 - HCF AND LCM	
	Use gcd-lcm relations with integer and positivity checks.	
+		+
	v	
+		+
	PANEL 05 - REMAINDERS	
	Reduce early and preserve the modulus.	
+		+
	v	
+		+
	PANEL 06 - UNIT DIGITS	
	Use cyclicity and treat exponent-zero cases separately.	
+		+
	v	
+		+
	PANEL 07 - DIGITS AND PLACE VALUE	
	Translate reversal and digit count into base-ten equations.	
+		+
	v	
+		+
	PANEL 08 - POWERS AND FACTORIALS	
	Use repeated division for prime exponents and trailing zeros.	
+		+
	v	
+		+
	PANEL 09 - FRACTIONS AND DECIMALS	
	Convert recurring forms through algebra, not memorised guesses.	
+		+
	v	
+		+
	PANEL 10 - SERIES	
	Test differences, ratios, alternation and position rules.	
+		+
	v	
+		+
	PANEL 11 - MAGNITUDE AND ESTIMATION	
	Bound before calculating exactly.	
+		+
	v	
+		+
	PANEL 12 - VERIFICATION	
	Plug back, check parity, digit count, remainder and order of magnitude.	

Number Systems and Number Sense - MUST-DO

Subject: CSAT | **Tier:** Must-Do (foundation) | **Family:** Basic numeracy / general mental ability. **Core skill:** solve by a **number property** (divisibility, unit digit, remainder, factor count) instead of heavy calculation. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). All drills below are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example (not a UPSC item). *Companion: Advanced Drill.*

1. Visual foundation



Core proposition: most CSAT number items are decided by a **last digit, a remainder, or a factor count** - computing the whole number is usually the slow, error-prone path.

2. Essential rules

Tool	Rule (exam-ready)	Edge condition
FACT: Div by 2 / 5 / 10	Look at the last digit (even; 0 or 5; 0).	-
FACT: Div by 4 / 8	Last 2 digits div by 4; last 3 digits div by 8.	-
FACT: Div by 3 / 9	Digit sum div by 3 / by 9.	Works in base 10 only.
FACT: Div by 11	(Sum of odd-place digits) - (sum of even-place digits) is 0 or a multiple of 11.	The difference may be negative ; judge its absolute value.
FACT: Unit-digit cycles	2,3,7,8 repeat with period 4; 4,9 with period 2; 0,1,5,6 are constant.	Applies to exponents $n \geq 1$; $a^0 = 1$ for any $a \neq 0$.
FACT: Trailing zeros of $n!$	$\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$ (count 5s).	Keep adding terms until one is 0.
FACT: Number of factors	For $n = p^a q^b r^c$, factor count = $(a+1)(b+1)(c+1)$.	p, q, r must be distinct primes ; the count includes 1 and n .
FACT: HCF x LCM	For two numbers, $\text{HCF} \times \text{LCM} = \text{product of the numbers}$.	Two numbers only, and both positive.

MEMORY: Unit-digit shortcut: for a power a^n , take $n \bmod 4$ (use 4 when it divides), then read the unit digit from a 's cycle. Example: $7^{100} \rightarrow 100 \bmod 4 = 0 \rightarrow$ last in cycle (7,9,3,1) $\rightarrow 1$.

3. Method (property-first)

- Decide whether the question wants the **exact value** or only a **property** (last digit, remainder, count, divisor).
- Pick the matching tool from Section 2.
- Reduce with **modular thinking**: replace big numbers by their remainders before combining.
- Only compute the full value if nothing else works - and then **estimate first** to catch slips.

4. Formulas and conditions

- FACT: $\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$, with $0 \leq \text{Remainder} < \text{Divisor}$.
- FACT: **Same-remainder divisor**: the greatest number dividing several numbers leaving the **same remainder** = HCF of their pairwise differences**.
- FACT: **Exact-division divisor**: greatest number dividing several numbers exactly = **HCF**; smallest number divisible by several = **LCM**.
- ANALYSIS: *Condition*: $\text{HCF} \times \text{LCM} = \text{product}$ holds for **two** numbers only, not three or more.
- ANALYSIS: *Condition*: a "remainder" is only well defined against a **stated divisor**, and it is always **less than** that divisor - an answer of "remainder 7 on division by 5" is arithmetically impossible.
- ANALYSIS: *Condition*: "a divides b" and "b divides a" are different claims; only one holds unless $a = b$. Divisibility is an **implication**, not an equivalence.

5. Core number sense methods

5.1 Recurring decimals

Form	Convert	Condition
$0.\overline{a}$	$a/9$	a is one recurring digit.
$0.\overline{ab}$	$ab/99$	ab is the two-digit repeating block.
$0.uv\overline{w}$	$(uvw - uv) / [10^2(10^{\text{len}(w)} - 1)]$	Subtract the non-repeating prefix; denominator has prefix zeros, then recurring-block 9s.

Worked example. $x = 0.1\overline{6}$. Then $100x = 16.\overline{6}$ and $10x = 1.\overline{6}$; subtract: $90x = 15$, so $x = 1/6$. Do not use the $a/9$ shortcut when a non-repeating prefix exists.

5.2 Place value, digit counts and reversal

- An n-digit positive integer lies from $10^{(n-1)}$ to $10^n - 1$.
- Digits used from 1 to N are counted in blocks: 1-9 uses 9×1 ; 10-99 uses 90×2 ; then count the final partial block. For a range L to U, use $\text{digits}(1..U) - \text{digits}(1..L-1)$.
- A two-digit number with digits a, b is $10a+b$; its reversal is $10b+a$. Their difference is $9(a-b)$ and their sum is $11(a+b)$. Here $a \neq 0$, and a reversed "two-digit number" also requires $b \neq 0$.

Worked example. Pages 1-250 use $9 + 90 \times 2 + 151 \times 3 = \mathbf{642 \text{ digits}}$. A number $10a+b$ minus its reversal is divisible by 9; this is a property, not proof that every multiple of 9 is a digit reversal.

5.3 Unknown digits and exponents

- For an unknown digit, apply the relevant divisibility test before trial: e.g. $47x2$ divisible by 9 requires $4+7+x+2 = 13+x$ divisible by 9, hence $x=5$.
- For non-zero a, $a^m a^n = a^{(m+n)}$, $a^m / a^n = a^{(m-n)}$, and $(a^m)^n = a^{(mn)}$. A negative exponent means reciprocal: $a^{(-n)} = 1/a^n$. Fractional exponents need a real-domain check: $a^{(1/2)}$ is real only for $a \geq 0$.
- For positive bases, powers increase with exponent when $a > 1$ and decrease when $0 < a < 1$.

Never raise an inequality to an even power or multiply it by an unknown-sign quantity without checking the direction and lost sign information.

Worked example. Compare $(1/2)^5$ and $(1/2)^3$: because the base lies between 0 and 1, the larger exponent gives the smaller value, so $(1/2)^5 < (1/2)^3$.

5.4 Counting multiples in a range

The count of multiples of positive d in the inclusive range $[L, U]$ is $\text{floor}(U/d) - \text{floor}((L-1)/d)$. For "divisible by a or b", add the two counts and subtract multiples of $\text{lcm}(a, b)$; "exactly one" subtracts the overlap twice.

Worked example. From 100 to 500, multiples of 6 number $83 - 16 = 67$, multiples of 10 number $50 - 9 = 41$, and multiples of 30 number $16 - 3 = 13$. Thus divisible by 6 **or** 10: $67 + 41 - 13 = 95$.

Routing boundary: Topic 02 owns number properties, digit structure and arithmetic counting. Cryptarithmic, code-language puzzles and general sequence inference are owned by Topic 06/08. Their appearances in appended PYQ ledgers are cross-owner demand references, not claims that this file independently teaches those separate mechanisms.

6. Original solved examples

□ Example A (unit digit)

Find the unit digit of 3^{35} . Cycle of 3 is (3, 9, 7, 1), period 4. $35 \bmod 4 = 3 \rightarrow$ third term $\rightarrow 7$. (Verified.)

□ Example B (remainder)

Remainder when 2^{50} is divided by 7. $2^3 = 8 \equiv 1 \pmod{7}$, period 3. $50 \bmod 3 = 2 \rightarrow 2^2 = 4$.
Remainder = 4. (Verified.)

□ Example C (HCF-LCM)

Two numbers have HCF 12 and LCM 180; one is 36. Find the other. Other = $(\text{HCF} \times \text{LCM}) / \text{known} = (12 \times 180) / 36 = 60$. (Verified.)

□ Example D (trailing zeros)

How many zeros end $25!$? $\text{floor}(25/5) + \text{floor}(25/25) = 5 + 1 = 6$. (Verified.)

□ Example E (factor count)

Number of factors of 360. $360 = 2^3 \times 3^2 \times 5^1 \rightarrow (3+1)(2+1)(1+1) = 24$. (Verified.)

7. Must-Know facts

- FACT: A number is div by 6 iff div by **2 and 3**; by 12 iff div by **3 and 4**. (Both work because the pairs are **coprime** - "div by 2 and 6" would **not** give div by 12.)
- FACT: Product of two numbers = HCF $\times$ LCM (two numbers only).
- FACT: Even $\times$ anything = even; odd $\times$ odd = odd; even + odd = odd (parity is a fast eliminator).
- FACT: Unit digit of any power a^n with $n \geq 1$ depends only on the **base's unit digit** and $n \bmod 4$ (taking a remainder of 0 as the 4th, last, cycle term).
- FACT: Trailing zeros of a factorial are governed by the count of **5s**, not 2s (2s are always more).
- FACT: A number has an **odd** number of factors exactly when it is a **perfect square**.

8. Common traps

- WRONG: Counting 2s for trailing zeros of $n!$. $\rightarrow$ Count **5s**; 2s are surplus.
- WRONG: Using $\text{HCF} \times \text{LCM} = \text{product}$ for **three** numbers. $\rightarrow$ Two numbers only.
- WRONG: Taking $n \bmod 4 = 0$ as "cycle position 0". $\rightarrow$ Use the **4th (last)** term of the cycle.
- WRONG: Forgetting the div-by-11 **alternating** sign. $\rightarrow$ It is odd-place minus even-place.
- WRONG: Splitting a divisor into non-coprime factors ("div by 2 and 6 $\Rightarrow$ div by 12"). $\rightarrow$ The factors must be **coprime**.
- WRONG: Confusing "same remainder" with "exact division". $\rightarrow$ Same remainder uses HCF of **differences**.

9. Quick checks

- FACT: Can you get a unit digit of any a^n in under 10 seconds via $n \bmod 4$?
- FACT: Can you state trailing zeros of $n!$ without multiplying?
- FACT: Given $n = p^a q^b$, can you write the factor count immediately?

10. Mini-drill (with answers and explanations)

1. Unit digit of 7^{100} .
2. Greatest number that divides 43, 91 and 183 leaving the **same** remainder.
3. Remainder when 15^{23} is divided by 4.
4. Number of zeros at the end of $100!$.
5. Next term: 2, 6, 12, 20, 30, ...

Answers.

1. **1.** Cycle (7,9,3,1); $100 \bmod 4 = 0 \rightarrow$ last term $\rightarrow 1$. (Verified.)
2. **4.** Differences 48, 92, 140; $\text{HCF}(48, 92, 140) = 4$. (Verified.)
3. **3.** $15 \equiv 3 \equiv -1 \pmod{4}$; $(-1)^{23} = -1 \equiv 3$. (Verified.)
4. **24.** $\text{floor}(100/5) + \text{floor}(100/25) = 20 + 4 = 24$. (Verified.)
5. **42.** Differences 4, 6, 8, 10 $\rightarrow$ next 12 $\rightarrow 30 + 12 = 42$ (these are $n(n+1)$). (Verified.)

11. Timed Core drill, diagnosis and retry gate

Set rule: attempt the eight items in **10 minutes** without a calculator. Before calculating, write the chosen method beside each answer: D divisibility, R remainder/cycle, F factors, G digits, E exponent, or I inclusion-exclusion.

1. Convert $0.\overline{2}$ to a fraction in lowest terms.
2. How many digits are used to number pages 1 to 120?
3. Find the missing digit in $53x4$ if it is divisible by 9.
4. How many multiples of 8 lie from 51 to 250 inclusive?
5. What is the unit digit of 7^{53} ?
6. How many numbers from 1 to 100 are divisible by exactly one of 4 and 6?
7. Which is larger: $(1/3)^4$ or $(1/3)^5$?
8. How many positive factors does $2^4 \times 3^2$ have?

Answers. 1. $5/18$ ($100x - 10x = 25$); 2. $9 + 180 + 63 = 252$; 3. $x = 6$;
4. $\text{floor}(250/8) - \text{floor}(50/8) = 31 - 6 = 25$; 5. **7**; 6. $25 + 16 - 2 \times 8 = 25$;
7. $(1/3)^4$; 8. $(4+1)(2+1) = 15$.

Miss pattern		Diagnosis			Mandatory repair before a new timed set				
Formula remembered but wrong base/block		C condition error			Rebuild one example from definitions, then solve three variants.				
Correct method, arithmetic slip		X execution			Estimate/order-check and rework the final two lines.				
Used a long calculation where a property worked		T tool selection			Label the property first on five easy items.				
Wrong "or/exactly one" count		A interpretation			Draw the overlap and write the lcm before counting.				
Date / set	Score / time	D	R	F	G	E	I	X / T	Next repair

Gate: proceed to optional Advanced work only after **7/8 in 10 minutes** on this set and **7/8 on a fresh set** after correcting misses. Otherwise repeat the failed method family; an Advanced remainder trick is not a substitute for Core number fluency. Apply the Master Framework's attempt and qualifying-margin rules in full-paper practice.

12. Study links

- FACT: Advanced companion - big-power remainders, highest prime powers, and inclusion-exclusion counting.
- FACT: Arithmetic and Commercial Math - ratio and percentage build on factor sense.
- FACT: Algebra, Inequalities and Data Sufficiency - number properties power many DS items.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2026.md. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** CSAT
- **Routed question demands:** 12

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	10	LCM multiple count in range	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	18	String-pattern position	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	19	Exponent equation factorisation	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	22	Measurement-combination optimisation	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	29	Discrete-set expression extrema	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	42	Unit digit of power product	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	45	Count of powers of two	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	49	Digit count in two-digit integers	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	57	Fibonacci-like recurrence	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	59	HCF-LCM cube relation	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	60	Divisibility-by-eleven digit puzzle	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	66	Reversed-digit divisibility count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- LCM multiple count in range
- String-pattern position
- Exponent equation factorisation
- Measurement-combination optimisation
- Discrete-set expression extrema
- Unit digit of power product
- Count of powers of two
- Digit count in two-digit integers
- Fibonacci-like recurrence
- HCF-LCM cube relation
- Divisibility-by-eleven digit puzzle
- Reversed-digit divisibility count

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** CSAT
- **Routed question demands:** 32

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	6	Operator-placement minimum	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	9	Trailing-zero count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	10	Cumulative savings sequence	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	15	Divisibility test	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	17	Terminal nonzero digit	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	18	Common-divisor remainder	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	19	HCF container sizing	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	20	Prime-sum unit digit	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	39	Expression parity	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	46	Periodic-sequence sum	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	54	Exponential divisibility	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	55	Integer-pair uniqueness	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	79	Number series	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	5	Factor-sum enumeration	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	6	Prime-progression sums	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	7	Unit-fraction solution count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	16	Derived-operator pattern	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	18	Simultaneous remainder constraints	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	20	Odd-product unit digit	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	25	Divisibility-exclusion count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	27	Prime-expression count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	28	LCM-HCF triple count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	35	Highest factor power	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	36	Number-series completion	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	40	Square-root digit count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	46	Power-sum remainder	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	47	Repdigit HCF	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	48	Concatenated-sequence digit	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	60	Number-series completion	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	64	Factor-count invariance	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	67	Power-series remainder	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	80	Interior-multiples count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Operator-placement minimum
- Trailing-zero count

- Cumulative savings sequence
- Divisibility test
- Terminal nonzero digit
- Common-divisor remainder
- HCF container sizing
- Prime-sum unit digit
- Expression parity
- Periodic-sequence sum
- Exponential divisibility
- Integer-pair uniqueness
- Number series
- Factor-sum enumeration
- Prime-progression sums
- Unit-fraction solution count
- Derived-operator pattern
- Simultaneous remainder constraints
- Odd-product unit digit
- Divisibility-exclusion count
- Prime-expression count
- LCM-HCF triple count
- Highest factor power
- Number-series completion
- Square-root digit count
- Power-sum remainder
- Repdigit HCF
- Concatenated-sequence digit
- Factor-count invariance
- Power-series remainder
- Interior-multiples count

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 65

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	3	Number matrix pattern	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	36	Digit-ordered number count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	9	Digit frequency count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	15	Divisibility set overlap	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	33	Reversed-digit ratio pairs	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	48	Number series pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	51	Number series pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	53	Page-digit range count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	60	Multi-digit divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	75	Digit sum divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	7	Trailing-zero count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	8	Cyclic-sum divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	10	Digit-sum primality	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	11	Digit symbol equation	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	12	HCF length measurement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	13	Prime-offset sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	14	Page-sum torn leaf	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	37	Digit and divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	50	Largest value among fractional and negative-exponent powers	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	51	HCF of measurements	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	53	Recurring decimal conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	54	LCM remainder constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	56	Product remainder calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	58	Digit sum of power	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	75	Difference-of-squares factoring	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	76	Fraction shift comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	77	Divisibility condition digit	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	5	Cyclic remainder pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	6	Divisibility digit value	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	26	Sequence incorrect term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	36	Digit-permutation sum divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	50	Number sequence missing term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	59	Cryptarithmic digit addition	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	60	Cryptarithmic digit multiplication	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	67	Consecutive-integer property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	76	Digit-interchange difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	79	Repunit smallest multiplier	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	4	Number sequence term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	5	Divisibility digit arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	9	Number magnitude comparison	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	47	LCM parity minimum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	54	Digit sum bound statements	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	58	Product modular remainder	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	65	LCM remainder threshold	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	66	Digit reversal product	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	67	Prime composite property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	74	Exponent maximum value	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	75	Number sequence term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	7	Product remainder modulo	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	8	Unit digit expansion	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	9	Digit-sum addition puzzle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	10	Digit-sum ratio minimum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	14	Integer parity determination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	15	Prime-composite property check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	16	Digit multiplication constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	17	Repeated-block divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	25	Conditional divisor count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	27	Repunit remainder modulo	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	28	Repunit square digit sum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	29	Range digit-sum total	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	45	Divisor-remainder bound count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	46	Repeated-digit sum constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	47	Permuted-digit number sum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	74	Large power remainder	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	76	Cryptarithmic digit solve	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Number matrix pattern
- Digit-ordered number count
- Digit frequency count
- Divisibility set overlap
- Reversed-digit ratio pairs
- Number series pattern
- Page-digit range count
- Multi-digit divisibility count
- Digit sum divisibility count
- Trailing-zero count
- Cyclic-sum divisibility
- Digit-sum primality
- Digit symbol equation
- HCF length measurement
- Prime-offset sequence
- Page-sum torn leaf
- Digit and divisibility count

- Largest value among fractional and negative-exponent powers
- HCF of measurements
- Recurring decimal conversion
- LCM remainder constraint
- Product remainder calculation
- Digit sum of power
- Difference-of-squares factoring
- Fraction shift comparison
- Divisibility condition digit
- Cyclic remainder pattern
- Divisibility digit value
- Sequence incorrect term
- Digit-permutation sum divisibility
- Number sequence missing term
- Cryptarithmic digit addition
- Cryptarithmic digit multiplication
- Consecutive-integer property
- Digit-interchange difference
- Repunit smallest multiplier
- Number sequence term
- Divisibility digit arrangement
- Number magnitude comparison
- LCM parity minimum
- Digit sum bound statements
- Product modular remainder
- LCM remainder threshold
- Digit reversal product
- Prime composite property
- Exponent maximum value
- Product remainder modulo
- Unit digit expansion
- Digit-sum addition puzzle
- Digit-sum ratio minimum
- Integer parity determination
- Prime-composite property check
- Digit multiplication constraint
- Repeated-block divisibility
- Conditional divisor count
- Repunit remainder modulo
- Repunit square digit sum
- Range digit-sum total
- Divisor-remainder bound count
- Repeated-digit sum constraint
- Permuted-digit number sum

- Large power remainder
- Cryptarithmic digit solve

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** Basic numeracy: numbers and their relations, orders of magnitude, etc. (Class X level).
- **Canonical scope:** Owns integer classes, divisibility, primes, factors, HCF/LCM, remainders, digits, powers, roots, factorial valuations, recurring decimals, magnitude and number series.
- **Cross-topic boundary:** Commercial percentages belong to Topic 03; rate contexts to Topic 04; algebraic unknowns and sufficiency to Topic 05; coding/counting structures to Topic 06.

Complete learner route

1. **Number universe:** Natural, whole, integer, rational, irrational and real numbers.
2. **Divisibility:** Use prime-factor and digit tests only within their valid base-ten conditions.
3. **Primes and factors:** Factorise before counting divisors or comparing powers.
4. **HCF and LCM:** Use gcd-lcm relations with integer and positivity checks.
5. **Remainders:** Reduce early and preserve the modulus.
6. **Unit digits:** Use cyclicity and treat exponent-zero cases separately.
7. **Digits and place value:** Translate reversal and digit count into base-ten equations.
8. **Powers and factorials:** Use repeated division for prime exponents and trailing zeros.
9. **Fractions and decimals:** Convert recurring forms through algebra, not memorised guesses.
10. **Series:** Test differences, ratios, alternation and position rules.
11. **Magnitude and estimation:** Bound before calculating exactly.
12. **Verification:** Plug back, check parity, digit count, remainder and order of magnitude.

Verification and hostile-query gate

Each shortcut is stated with its modulus, parity, cycle or factorisation condition. Deterministic checks recompute every generated answer by direct arithmetic or enumeration.

The hostile absence search explicitly tested these families and close-option terms: **divisibility; prime; factor; HCF; LCM; remainder; unit digit; trailing zeros**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess; return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Practice variant 1: What is the unit digit of 2^{11} ?

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q2. Practice variant 1: What is the remainder when 137 is divided by 7?

A. 5 B. 4 C. 3 D. 6

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q3. Practice variant 1: What is the HCF of 12 and 18?

A. 7 B. 5 C. 6 D. 8

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q4. Practice variant 1: How many positive divisors does 60 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q5. Practice variant 1: How many trailing zeros are in $25!$?

A. 6 B. 7 C. 5 D. 8

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q6. Practice variant 1: How many multiples of 7 lie from 10 through 100, inclusive?

A. 14 B. 13 C. 12 D. 15

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q7. Practice variant 1: The sequence starts 3, 5, 7, What is its 8th term?

A. 18 B. 16 C. 17 D. 19

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q8. Practice variant 1: How many decimal digits are in 137 ?

A. 4 B. 2 C. 5 D. 3

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Q9. Practice variant 2: What is the unit digit of 3^{12} ?

A. 1 B. 2 C. 0 D. 3

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q10. Practice variant 2: What is the remainder when 154 is divided by 8?

A. 3 B. 2 C. 1 D. 4

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q11. Practice variant 2: What is the HCF of 14 and 21?

A. 8 B. 6 C. 7 D. 9

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q12. Practice variant 2: How many positive divisors does 72 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q13. Practice variant 2: How many trailing zeros are in $30!$?

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q14. Practice variant 2: How many multiples of 8 lie from 11 through 105, inclusive?

A. 13 B. 12 C. 11 D. 14

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q15. Practice variant 2: The sequence starts 4, 7, 10, What is its 8th term?

A. 26 B. 24 C. 25 D. 27

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q16. Practice variant 2: How many decimal digits are in 1037^n ?

A. 5 B. 3 C. 6 D. 4

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	6	Operator-placement minimum	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	9	Trailing-zero count	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	10	Cumulative savings sequence	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	15	Divisibility test	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	17	Terminal nonzero digit	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	18	Common-divisor remainder	C (supplied)	Translate to congruences and plug the result into every condition.
2024	19	HCF container sizing	B (supplied)	Use prime factors or Euclid; verify the common-divisor condition.
2024	20	Prime-sum unit digit	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	39	Expression parity	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	46	Periodic-sequence sum	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	54	Exponential divisibility	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	55	Integer-pair uniqueness	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	79	Number series	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	5	Factor-sum enumeration	C (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	6	Prime-progression sums	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	7	Unit-fraction solution count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	16	Derived-operator pattern	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	18	Simultaneous remainder constraints	C (supplied)	Translate to congruences and plug the result into every condition.
2025	20	Odd-product unit digit	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	25	Divisibility-exclusion count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	27	Prime-expression count	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	28	LCM-HCF triple count	D (supplied)	Use prime factors or Euclid; verify the common-divisor condition.
2025	35	Highest factor power	B (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	36	Number-series completion	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	40	Square-root digit count	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	46	Power-sum remainder	C (supplied)	Translate to congruences and plug the result into every condition.
2025	47	Repdigit HCF	C (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	48	Concatenated-sequence digit	D (supplied)	Use place value, cyclicity or a constrained enumeration.

Year	Q	Neutral verified demand	Key status	Solution architecture
2025	60	Number-series completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	64	Factor-count invariance	C (supplied)	Prime-factorise, count exponents and verify divisibility.
2025	67	Power-series remainder	A (supplied)	Translate to congruences and plug the result into every condition.
2025	80	Interior-multiples count	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	10	LCM multiple count in range	D (provisional)	Use prime powers and range bounds.
2026	18	String-pattern position	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	19	Exponent equation factorisation	B (provisional)	Prime-factorise, count exponents and verify divisibility.
2026	22	Measurement-combination optimisation	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	29	Discrete-set expression extrema	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	42	Unit digit of power product	C (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	45	Count of powers of two	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	49	Digit count in two-digit integers	B (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	57	Fibonacci-like recurrence	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	59	HCF-LCM cube relation	C (provisional)	Use prime factors or Euclid; verify the common-divisor condition.
2026	60	Divisibility-by-eleven digit puzzle	D (provisional)	Use place value, cyclicity or a constrained enumeration.
2026	66	Reversed-digit divisibility count	B (provisional)	Use place value, cyclicity or a constrained enumeration.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. Practice variant 5: What is the unit digit of 6^{15} ?

A. 6 B. 7 C. 5 D. 8

Correct answer: A. Reduce powers modulo 10 and use the complete unit-digit cycle.

Q34. Practice variant 5: What is the remainder when 205 is divided by 11?

A. 8 B. 7 C. 6 D. 9

Correct answer: B. Division algorithm: $n = mq + r$ with $0 \leq r < m$.

Q35. Practice variant 5: What is the HCF of 20 and 30?

A. 11 B. 9 C. 10 D. 12

Correct answer: C. Prime factorisation or Euclid's algorithm gives the greatest common divisor.

Q36. Practice variant 5: How many positive divisors does 108 have?

A. 13 B. 11 C. 14 D. 12

Correct answer: D. If $n = \text{product } p_i^{a_i}$, the divisor count is $\text{product } (a_i + 1)$.

Q37. Practice variant 5: How many trailing zeros are in $45!$?

A. 10 B. 11 C. 9 D. 12

Correct answer: A. Count factors of 5 by repeated division; factors of 2 are more abundant.

Q38. Practice variant 5: How many multiples of 11 lie from 14 through 120, inclusive?

A. 10 B. 9 C. 8 D. 11

Correct answer: B. Use $\text{floor}(\text{high}/d) - \text{floor}((\text{low}-1)/d)$.

Q39. Practice variant 5: The sequence starts 7, 13, 19, What is its 8th term?

A. 50 B. 48 C. 49 D. 51

Correct answer: C. This is an arithmetic progression: $a_n = a + (n-1)d$.

Q40. Practice variant 5: How many decimal digits are in 1000037?

A. 8 B. 6 C. 9 D. 7

Correct answer: D. A positive integer n has $\text{floor}(\log_{10} n) + 1$ digits; direct place-value bounding gives the same answer.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Number Systems and Number Sense - ADVANCED

Subject: CSAT | **Tier:** Advanced Drill | **Family:** Basic numeracy / general mental ability. **Core skill:** big-power remainders, highest prime powers in factorials, last-two-digit cycles, and inclusion-exclusion counting - all by property, at speed. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example. *Companion: Foundation.*

1. Architecture

```
BIG NUMBER / BIG POWER
|
list  $a^n \bmod m$  (locate any pre-period, then the period)
|
tower? -> first prove exponent reaches periodic segment
         then map it within that segment; otherwise evaluate directly
|
counting how many? -> floor-division + inclusion-exclusion
|
ANSWER without ever forming the giant number
```

Analytical claim: large-power items become manageable after reducing the base modulo m and then checking whether the residue sequence is periodic immediately, has a pre-period, or stabilizes. Exponent reduction modulo a period is valid only from the point where that period actually applies.

2. Advanced tools with conditions

Tool	Statement	Condition / caution								
FACT: Cyclic remainders	If the powers are periodic from the relevant exponent onward with period k , reduce the exponent within that periodic segment.	List powers first. When $\gcd(a, m) \neq 1$, check for a pre-period or stabilization; e.g. $2^n \bmod 8$ becomes 0 from $n = 3$.								
FACT: Power tower	For $a^{(b^c)} \bmod m$, reduce the exponent modulo a verified period only after accounting for any pre-period.	If $\gcd(a, m) = 1$, Euler/Carmichael-style periodicity is available; otherwise inspect the residue sequence directly.								
FACT: Exponent of prime p in $n!$	$v_p(n!) = \lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \dots$; the highest power dividing $n!$ is $p^{v_p(n!)}$.	Trailing zeros equal $v_5(n!)$ (since 2s are more plentiful).								
FACT: Last two digits	Track $\bmod 100$; many bases have short 2-digit cycles.	e.g., $7^4 \equiv 01 \pmod{100}$.								
FACT: Inclusion-exclusion	$\setminus$	$A \sqcup B \setminus$	$= \setminus$	$A \setminus$	$+ \setminus$	$B \setminus$	$- \setminus$	$A \cap B \setminus$	$\setminus$	For "divisible by neither", subtract the union from the range.

3. Harder methods (worked)

□ Method 1 - power tower remainder

Remainder of $32^{(32^{32})}$ divided by 7. $32 \equiv 4 \pmod{7}$; powers of 4 mod 7 cycle (4, 2, 1) with period 3.

Reduce exponent: $32 \equiv 2 \pmod{3}$, so $32^{32} \equiv 2^{32} \equiv (-1)^{32} \equiv 1 \pmod{3}$. Hence $4^1 = 4$. **Remainder = 4.** (Verified.)

□ Method 2 - exponent of a prime in a factorial

Exponent of 3 in $100!$. $\lfloor 100/3 \rfloor + \lfloor 100/9 \rfloor + \lfloor 100/27 \rfloor + \lfloor 100/81 \rfloor = 33 + 11 + 3 + 1 = 48$, so the highest power of 3 dividing $100!$ is 3^{48} . (Verified.)

□ Method 3 - last two digits

Last two digits of 7^{100} . $7^4 = 2401 \equiv 01 \pmod{100}$, so $7^{100} = (7^4)^{25} \equiv 01$. **Last two digits = 01.** (Verified.)

□ Method 4 - when the base and modulus share a factor (find the eventual period)

ANALYSIS: The "reduce the exponent modulo the cycle length" habit **fails** when $\gcd(a, m) \neq 1$. List the residues first:

n	1	2	3	4	5	6
$6^n \bmod 8$	6	4	0	0	0	0
$2^n \bmod 8$	2	4	0	0	0	0

The sequence has a **pre-period** and then stabilises at 0 from $n = 3$ onward (an eventual period of 1), because $8 = 2^3$ divides every 6^n and 2^n with $n \geq 3$. So $6^{25} \bmod 8 = 0$. Reducing 25 by a cycle assumed to start at exponent 1 is unsafe; first locate the pre-period and eventual period. (Verified.)

- **MEMORY: Escalation rule:** before reducing an exponent, ask "**is** $\gcd(\text{base}, \text{modulus}) = 1$?" If yes, powers are periodic from exponent 1; verify the cycle length before reducing. If no, write out residues and locate any **pre-period plus eventual cycle** (which may be a constant); non-coprime bases can still cycle immediately.

4. Time-saving techniques (safe conditions)

- ANALYSIS: **Coprimality gate** - run this before any exponent reduction. If $\gcd(\text{base}, \text{modulus}) = 1$, a genuine cycle exists and reducing the exponent modulo its length is safe. If not, list residues and look for a **pre-period or stabilisation** instead (Method 4). *This single check prevents the most expensive error in the family.*
- ANALYSIS: **Negatives modulo m**. Replace a base by a small negative residue (e.g., $15 \equiv -1 \pmod{4}$) to make powers trivial. *Safe always.*
- ANALYSIS: **Parity/last-digit elimination**. Kill options of the wrong parity or unit digit before any full solve. *Safe always.*
- ANALYSIS: **Count 5s for factorial zeros**. *Safe always (2s dominate).*
- ANALYSIS: **Exponent reduction requires a verified periodic segment**; list powers first. **Do not assume immediate periodicity or a universal period, especially when the base and modulus are non-coprime.**

5. Boundary cases

- ANALYSIS: Within a cycle that starts at exponent 1, a reduced exponent of 0 selects the ****last cycle position****. If there is a pre-period, map the exponent relative to the cycle's actual start instead.
- ANALYSIS: A **stabilising** sequence has an eventual constant cycle (period 1). Do not apply a period assumed from exponent 1: first check that the exponent lies beyond its pre-period.
- ANALYSIS: "Divisible by **either**" ($|A| + |B| - |A \cap B|$) and "divisible by **exactly one**" ($|A| + |B| - 2|A \cap B|$) are different counts; read which is asked.
- ANALYSIS: Inclusion-exclusion needs the **intersection** term (divisible by lcm), not the product of the two divisors, unless they are coprime.
- ANALYSIS: $\text{HCF} \times \text{LCM} = \text{product}$ **fails** for three or more numbers - use prime factorisation instead.
- ANALYSIS: A "last two digits" answer below 10 keeps its **leading zero** ($3^{101} \rightarrow 03$, not 3).

6. Advanced traps

- WRONG: Reducing the exponent modulo m instead of modulo the **cycle length**. -> Reduce by the period.
- WRONG: Reducing modulo a period before checking non-coprime stabilization/pre-period. -> List residues and locate where repetition actually begins.
- WRONG: Counting trailing zeros of $n!$ by 2s. -> Count 5s.
- WRONG: Stopping the $v_p(n!)$ floor sum one term early. -> Continue until the term is 0.
- WRONG: In "divisible by neither 2 nor 3", subtracting $50 + 33$ without adding back multiples of 6. -> Use inclusion-exclusion.
- WRONG: Answering the "either" count when "exactly one" was asked. -> Subtract the overlap **twice**.
- WRONG: Treating a^0 inside a cycle as term 0. -> It is the last cycle term.

7. Error analysis

Recurring miss	Master-Framework code	Fix
Wrong cycle length	C	List powers until repeat before reducing.
Off-by-one in floor sums	X	Re-add the floor terms; check the last non-zero term.
Forgetting intersection term	A	Write the inclusion-exclusion formula every time.
Slips on giant multiplications	X	Never form the giant number; stay in mod.

8. Advanced drill (with full solutions)

ANALYSIS: These items are **new** - none repeats Methods 1-4 above or an example in the Foundation file.

1. Trailing zeros of $50!$.
2. How many integers from 1 to 1000 are divisible by **3 or 5**?
3. Remainder when 2^{100} is divided by 9.
4. What is the **highest power of 2** that divides $50!$?
5. Last two digits of 3^{101} .
6. How many integers from 1 to 300 are divisible by **exactly one** of 4 and 6?
7. Remainder when 10^{30} is divided by 16.

Solutions.

1. **12.** $\text{floor}(50/5) + \text{floor}(50/25) = 10 + 2 = 12.$ (Verified.)
2. **467.** $\text{floor}(1000/3) + \text{floor}(1000/5) - \text{floor}(1000/15) = 333 + 200 - 66 = 467.$ (Verified.)
3. **7.** $\text{gcd}(2, 9) = 1$, so a cycle exists: $2^6 = 64 \equiv 1 \pmod{9}$, period **6**.
 $100 \pmod{6} = 4 \rightarrow 2^4 = 16 \equiv 7.$ (Verified.)
4. $2^{47} \cdot v_2(50!) = 25 + 12 + 6 + 3 + 1 = 47.$ *(Verified - keep adding floors until the term is 0; the last non-zero term here is $\text{floor}(50/32) = 1.$)*
5. **03.** $3^{20} \equiv 01 \pmod{100}$, so $3^{100} \equiv 01$ and $3^{101} \equiv 3$, i.e. the last two digits are **03** - write the leading zero. (Verified.)
6. **75.** $|A| = \text{floor}(300/4) = 75$, $|B| = \text{floor}(300/6) = 50$, $|A \cap B| = \text{floor}(300/12) = 25.$
"Exactly one" = $|A| + |B| - 2|A \cap B| = 75 + 50 - 50 = 75.$ (Verified.) ANALYSIS: Subtracting the overlap **once** (the "either" count) would give 100 - a different question.
7. **0.** $\text{gcd}(10, 16) = 2 \neq 1$, so there is **no cycle to reduce into** - apply the Method 4 gate.
Listing residues gives 10, 4, 8, 0, 0, ...: the sequence **stabilises** at 0 from $n = 4$, because $16 = 2^4$ divides 10^n once $n \geq 4$. Since $30 \geq 4$, the remainder is **0**. (Verified.)

9. PYQ-pattern notes (2024-2026, Set A)

- ANALYSIS: Number-system items are a **substantial recurring block**: 13/80 in 2024, 19/80 in 2025, and a provisional 12/80 in 2026. They reward the property-first habit far more than long division.

- ANALYSIS: Recurring shapes: **unit-digit / last-digit of a power, remainder of a large power, HCF/LCM word problems, trailing-zeros / factor-count, and number/letter series.**

- ANALYSIS: The papers reward candidates who **reduce before computing**; brute force here is the classic time sink flagged in the Master Framework.

10. Study links

- FACT: Foundation companion.
- FACT: Algebra, Inequalities and Data Sufficiency - number properties decide many data-sufficiency items.
- FACT: Logical Reasoning, Coding, Counting and DI - counting/inclusion-exclusion overlap.

Ownership note: appended historical ledgers may list cryptarithmic or broad sequence demands beside number questions. Their mechanism is owned by Topic 06/08; this companion supplies only number-property prerequisites and must not be used to relabel those topics as Number Systems.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 65

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	3	Number matrix pattern	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	36	Digit-ordered number count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	9	Digit frequency count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	15	Divisibility set overlap	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	33	Reversed-digit ratio pairs	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	48	Number series pattern	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 08 for sequence method; retain Topic 02 only for numeric properties.
2019	CSAT	51	Number series pattern	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 08 for sequence method; retain Topic 02 only for numeric properties.
2019	CSAT	53	Page-digit range count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	60	Multi-digit divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	75	Digit sum divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	7	Trailing-zero count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	8	Cyclic-sum divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	10	Digit-sum primality	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	11	Digit symbol equation	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	12	HCF length measurement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	13	Prime-offset sequence	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	14	Page-sum torn leaf	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	37	Digit and divisibility count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	50	Largest value among fractional and negative-exponent powers	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	51	HCF of measurements	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	53	Recurring decimal conversion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	54	LCM remainder constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	56	Product remainder calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	58	Digit sum of power	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	75	Difference-of-squares factoring	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	76	Fraction shift comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	77	Divisibility condition digit	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	5	Cyclic remainder pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	6	Divisibility digit value	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	26	Sequence incorrect term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	36	Digit-permutation sum divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	50	Number sequence missing term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	59	Cryptarithmic digit addition	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 06 for the puzzle mechanism; Topic 02 supplies digit constraints only.
2021	CSAT	60	Cryptarithmic digit multiplication	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 06 for the puzzle mechanism; Topic 02 supplies digit constraints only.
2021	CSAT	67	Consecutive-integer property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	76	Digit-interchange difference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	79	Repunit smallest multiplier	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	4	Number sequence term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	5	Divisibility digit arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	9	Number magnitude comparison	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	47	LCM parity minimum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	54	Digit sum bound statements	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	58	Product modular remainder	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	65	LCM remainder threshold	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	66	Digit reversal product	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	67	Prime composite property	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	74	Exponent maximum value	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	75	Number sequence term	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	7	Product remainder modulo	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	8	Unit digit expansion	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	9	Digit-sum addition puzzle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	10	Digit-sum ratio minimum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	14	Integer parity determination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	15	Prime-composite property check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	16	Digit multiplication constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	17	Repeated-block divisibility	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	25	Conditional divisor count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	27	Repunit remainder modulo	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	28	Repunit square digit sum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	29	Range digit-sum total	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	45	Divisor-remainder bound count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	46	Repeated-digit sum constraint	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	47	Permuted-digit number sum	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	74	Large power remainder	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	76	Cryptarithmic digit solve	Objective question; official key unavailable locally	Cross-owner reference; key unavailable locally	Route to Topic 06 for the puzzle mechanism; Topic 02 supplies digit constraints only.

What this owner must now support

- Number matrix pattern
- Digit-ordered number count
- Digit frequency count
- Divisibility set overlap
- Reversed-digit ratio pairs

- Number series pattern - routed to Topic 08 (cross-owner reference)
- Page-digit range count
- Multi-digit divisibility count
- Digit sum divisibility count
- Trailing-zero count
- Cyclic-sum divisibility
- Digit-sum primality
- Digit symbol equation
- HCF length measurement
- Prime-offset sequence
- Page-sum torn leaf
- Digit and divisibility count
- Largest value among fractional and negative-exponent powers
- HCF of measurements
- Recurring decimal conversion
- LCM remainder constraint
- Product remainder calculation
- Digit sum of power
- Difference-of-squares factoring
- Fraction shift comparison
- Divisibility condition digit
- Cyclic remainder pattern
- Divisibility digit value
- Sequence incorrect term - routed to Topic 08 (cross-owner reference)
- Digit-permutation sum divisibility
- Number sequence missing term - routed to Topic 08 (cross-owner reference)
- Cryptarithmic digit addition - routed to Topic 06 (cross-owner reference)
- Cryptarithmic digit multiplication - routed to Topic 06 (cross-owner reference)
- Consecutive-integer property
- Digit-interchange difference
- Repunit smallest multiplier
- Number sequence term - routed to Topic 08 (cross-owner reference)
- Divisibility digit arrangement
- Number magnitude comparison
- LCM parity minimum
- Digit sum bound statements
- Product modular remainder
- LCM remainder threshold
- Digit reversal product
- Prime composite property
- Exponent maximum value
- Product remainder modulo
- Unit digit expansion
- Digit-sum addition puzzle

- Digit-sum ratio minimum
- Integer parity determination
- Prime-composite property check
- Digit multiplication constraint
- Repeated-block divisibility
- Conditional divisor count
- Repunit remainder modulo
- Repunit square digit sum
- Range digit-sum total
- Divisor-remainder bound count
- Repeated-digit sum constraint
- Permuted-digit number sum
- Large power remainder
- Cryptarithmic digit solve - routed to Topic 06 (cross-owner reference)

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Number universe:** Natural, whole, integer, rational, irrational and real numbers.
- **Divisibility:** Use prime-factor and digit tests only within their valid base-ten conditions.
- **Primes and factors:** Factorise before counting divisors or comparing powers.
- **HCF and LCM:** Use gcd-lcm relations with integer and positivity checks.
- **Remainders:** Reduce early and preserve the modulus.
- **Unit digits:** Use cyclicity and treat exponent-zero cases separately.
- **Digits and place value:** Translate reversal and digit count into base-ten equations.
- **Powers and factorials:** Use repeated division for prime exponents and trailing zeros.
- **Fractions and decimals:** Convert recurring forms through algebra, not memorised guesses.
- **Series:** Test differences, ratios, alternation and position rules.
- **Magnitude and estimation:** Bound before calculating exactly.
- **Verification:** Plug back, check parity, digit count, remainder and order of magnitude.

Ownership and close-option firewall

- **Own here:** Owns integer classes, divisibility, primes, factors, HCF/LCM, remainders, digits, powers, roots, factorial valuations, recurring decimals, magnitude and number series.
- **Do not duplicate:** Commercial percentages belong to Topic 03; rate contexts to Topic 04; algebraic unknowns and sufficiency to Topic 05; coding/counting structures to Topic 06.
- **Verification:** Each shortcut is stated with its modulus, parity, cycle or factorisation condition. Deterministic checks recompute every generated answer by direct arithmetic or enumeration.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.